Summary:

This paper builds a general equilibrium incomplete markets model with housing choice and drug use/addiction. The author uses the model to measure the effect of different rental voucher policies on homelessness and drug use/addiction. The baseline policy is a rent voucher program that is unconditional on drug-use status, but rationed: only a fraction of income-eligible people receive vouchers. There are three alternative policies. One excludes drug users, but still rations the vouchers among those who are income and drug-use eligible. Then there are policies that drop the rationing of vouchers, with and without the drug-user exclusion. The long-run social value of these policies rests on how they change the rate of drug use and addiction. As such, the policy that provides a rental voucher to everyone who is drug free and income eligible delivers the highest welfare gain. It provides valuable insurance against a very bad outcome (homelessness) contingent on being drug-free; people know that if they start experimenting with drugs then they run a risk of becoming addicted and therefore will have a harder time quitting to get a voucher.

Conceptual Concerns

This is a very rich model that builds significantly (and I believe describes more clearly) the framework from Imrohoroglu and Zhao. However, all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access. The only empirical drug use elasticity mentioned is a price elasticity of drug use participation, but it isn't clear to me that this provides enough or the correct discipline for these policy experiments. I would like to see that the model matches some evidence on the response of drug use to housing access (see Milby, et al (2005) for cocaine users, but there are many papers on this topic.)

When I think about the welfare effects of drug use and homelessness, there are two margins that this paper misses. First, a large homeless population, especially if many are addicted to drugs, creates a negative externality beyond the fiscal burden of providing services. Second, parents suffer anxiety that their children may become drug addicts and/or homeless in adulthood. One structural rental housing paper that considers the negative externality of homelessness (though doesn't have drug addiction) is Corbae, et al (2024). I do not know of a paper that includes drug addiction in a dynastic overlapping generations framework, but as soon as parents value the well being of their children this additional welfare consideration would arise. Of these two, the externality seems very easy to introduce into the current model (though would require a new calibration moment), while the dynastic element seems like more of a stretch, but could be discussed (I think it would only amplify the welfare gains from reducing addiction).

The cost of having such a rich model is that it can be a bit of a black box. It would be helpful to have a section where the mechanisms of the model are explored more thoroughly. For example, it would be useful to show and explain intuitively a plot of the drug use policy function for addicts and non-addicts as their income and wealth vary (for various reasonable values of other states). It would also help to further explain the importance of the period utility specification in (13) which, I think, implies that drug *addiction* is more attractive for low income people (who have low values of non-drug consumption and housing). Currently, the

discussion on that page is all about how this function affects the value of drug use, but it also bakes in an income-contingent gain/loss from becoming addicted by shifting the utility weight towards drug consumption away from non-drug consumption and housing.

Finally, it would be interesting to know a bit more about optimal housing voucher programs in this model. The parameter χ_H governs the fiscal cost of having somebody unhoused and appears pretty high at 25% of average income. Would there be significant further welfare gains from broadening the voucher program by making higher incomes eligible or covering more of the rent (with the drug-free contingency)? My (perhaps wrong) reading is that the cost of having somebody unhoused is significantly higher than the rent on the smallest housing unit.

Detail concerns:

I will not nitpick since there would be no end to how much robustness could be done with a model this complex. Furthermore, the paper describes most things quite well. The only thing I am unsure about is the housing grid. It has seven values but only three are calibrated (the smallest rental size, the value of being homeless, and the smallest home size for purchase). The full grid is then listed in footnote 4 rather than a calibration table. My assumption before seeing the grid was that the bounds were calibrated and then the sizes were distributed uniformly (which would be some justification not to treat each size as a free parameter). But they are not uniformly spread out, at least in levels or ratios. So, how are the housing levels being chosen and does it matter for results? Could they just be spread uniformly without affecting the calibration and policy conclusions? If not then I think they need to be included as parameters in the calibration.

Papers Cited Here Not in Paper

Milby, J.B., Schumacher, J.E., Wallace, D., Freedman, M.J. and Vuchinich, R.E., 2005. To house or not to house: The effects of providing housing to homeless substance abusers in treatment. American journal of public health, 95(7), pp.1259-1265.

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